

Stochastic Simulation Analysis: Waste Collection Model

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1 State Representation

The system state at any time $t \geq 0$ is represented by the state vector

$$\mathcal{S}(t) = (Q_1(t), Q_2(t), Q_3(t), T_1(t), T_2(t), T_3(t)),$$

where $Q_i(t) \in \mathbb{N}_0$ denotes the number of waste-collection requests currently present in district $i \in \{1, 2, 3\}$, including both waiting and in-service requests, and $T_j(t)$ describes the status of truck $j \in \{1, 2, 3\}$. Each truck state records whether the truck is idle or busy, the district currently being served, and the waste type associated with the active service.

2 Event logic

The discrete-event simulation updates the system state and Future Event List (FEL) through three event types.

Arrivals Requests arrive at district $i \in \{1, 2, 3\}$ via independent Poisson processes with rates λ_i . Each arrival is independently assigned a waste type: Type 1 with probability q_1 , Type 2 with q_2 , or Type 3 with $1 - q_1 - q_2$. Requests are appended to their district queue. If the home truck is idle, service begins and a Service Completion Event is scheduled. Otherwise, other trucks are checked for idleness in descending order. Service times S depend on the waste type:

$$S_1 \sim \text{Erlang}(k_1, \mu_1), \quad S_2 \sim \text{Erlang}(k_2, \mu_2), \quad S_3 \sim \text{Exp}(\mu_3).$$

If a request arrives at the home district of a truck currently serving a foreign district, a Rerouting Event is triggered if the updated home queue length exceeds $K = 5$.

Service Completions Upon completing a service, the district's completion counter gets increased. The truck first checks its home queue. If non-empty, the head-of-line request enters service (FCFS) and a new completion event is scheduled. If empty, the truck inspects foreign districts using a fixed order:

$$\text{Truck 1: } 2 \rightarrow 3, \quad \text{Truck 2: } 3 \rightarrow 1, \quad \text{Truck 3: } 1 \rightarrow 2.$$

At each non-empty foreign queue, a Bernoulli trial with success probability p determines if the truck accepts the request. On success, it begins service immediately. On failure, it moves to the next district. If all options are empty or rejected, the truck idles at home.

Reroutings A Rerouting Event can only happen to a truck serving a foreign district when its home queue exceeds $K = 5$. The interrupted request returns to the head of its original queue and its partial service is discarded, meaning service will restart from scratch with a new sampled time. Assuming travel times are negligible, the truck instantaneously returns home to service the oldest request, scheduling a Service Completion Event and increasing the rerouting counter.

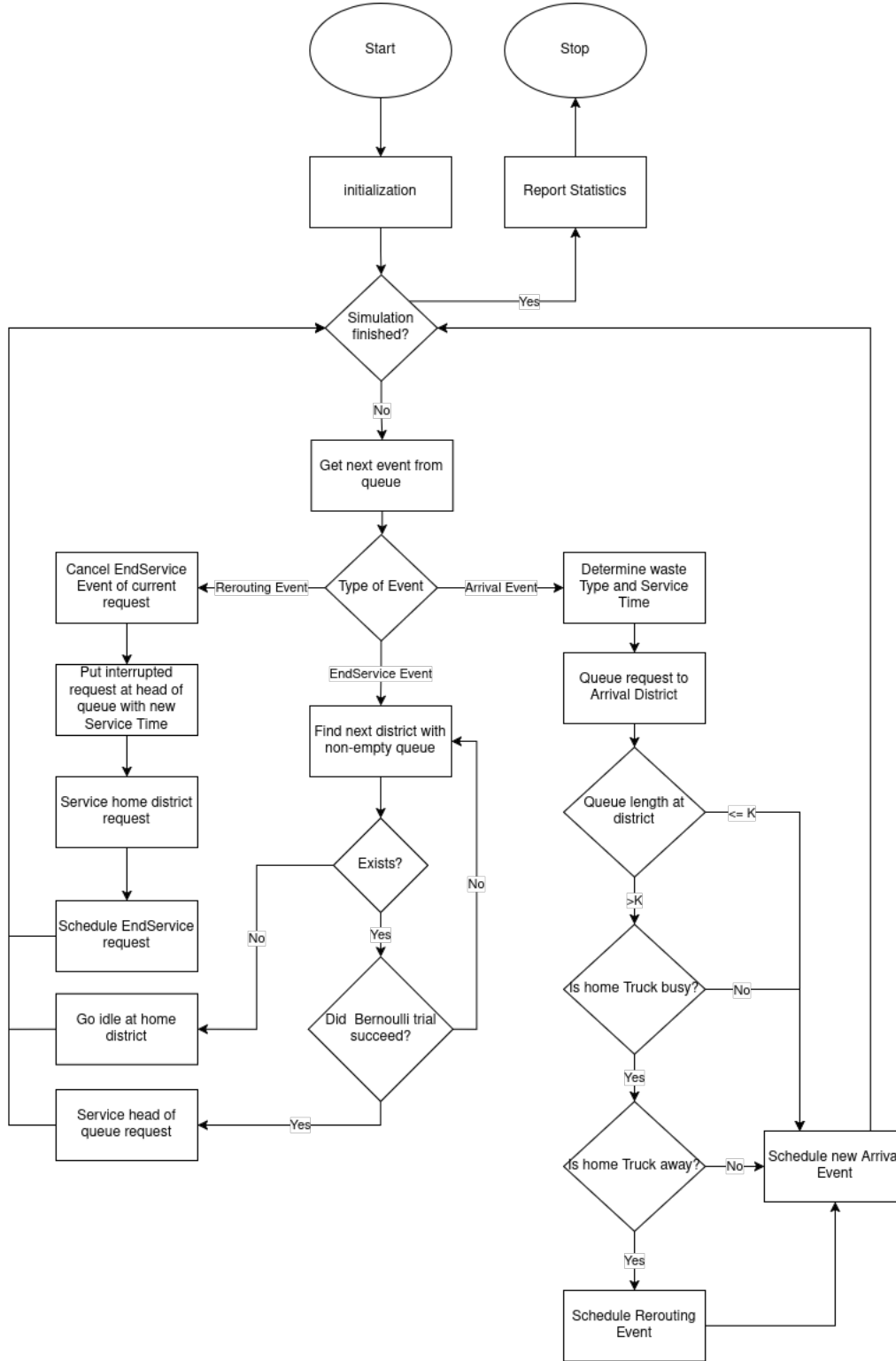


Figure 1: Flowchart of the Waste Collection Simulation Model

3 Verification Cases

To verify the correctness of the model, we consider simplified settings for which closed-form analytical results are available. We then compare these theoretical benchmarks with simulation outputs. Two representative verification cases are presented below.

3.1 V1 (M/M/1 reduction)

We first consider the special case where $N = M = 1$ (with N denoting the number of trucks and M the number of districts), $p = 0$ (no friendliness effect), and only Type 3 service is active with exponential service times $\text{Exp}(\mu_3)$. Arrivals follow a Poisson process with rate λ_1 . Under these assumptions, the system reduces to a standard $M/M/1$ queue. The theoretical steady-state mean waiting time is given by

$$\mathbb{E}[W] = \frac{1}{\mu_3 - \lambda_1}, \quad \lambda_1 < \mu_3. \quad (1)$$

For validation, we set $\lambda_1 = 0.5$ and $\mu_3 = 1.0$, yielding $\mathbb{E}[W] = 2$. Table 1 compares this value with simulation results for increasing run lengths.

End Time	W_{sim}	$ \mathbb{E}[W] - W_{\text{sim}} $
10^1	3.596	1.596
10^2	2.395	0.395
10^3	2.122	0.122
10^4	2.055	0.055
10^5	2.029	0.029
10^6	2.006	0.006
10^7	2.001	0.001

Table 1: Simulation results for the $M/M/1$ reduction with $\lambda_1 = 0.5$ and $\mu_3 = 1.0$, evaluated over increasing simulation horizons.

The results show that as the simulation run length increases, the simulated mean waiting time converges to the theoretical value. While short runs are highly variable, longer runs reduce the error to a negligible amount (around 10^{-6} to 10^{-7}). This confirms that the model shows $M/M/1$ queue behaviour.

3.2 V2 (M/M/2 reduction)

Next, we consider the symmetric case $N = M = 2$, $p = 1$, with equal arrival rates $\lambda_1 = \lambda_2 = \lambda$ and equal service rates $\mu_1 = \mu_2 = \mu$, restricted to Type 3 service only. Rerouting is disabled (e.g., $K = \infty$) so that no threshold effects occur. Under full symmetry and complete pooling, the system reduces to an $M/M/2$ queue. The total mean waiting time in queue is given by

$$\mathbb{E}[W] = \frac{C(2, \frac{2\lambda}{\mu})}{2\mu - 2\lambda} + \frac{1}{\mu}. \quad (2)$$

Here, $C(2, a)$ denotes the Erlang–C delay probability. For two servers, this simplifies to $C(2, a) = \frac{a^2}{2+a}$. For the numerical experiment, we choose $\lambda = 0.6$ and $\mu = 1.0$. Hence $\mathbb{E}[W] = \frac{C(2, 1.2)}{0.8} + 1 = 1.5625$. Table 2 compares this theoretical value with simulation results.

End Time	W_{sim}	$ \mathbb{E}[W] - W_{\text{sim}} $
10^1	2.863	1.301
10^2	2.426	0.864
10^3	1.948	0.386
10^4	1.545	0.018
10^5	1.560	0.003
10^6	1.561	0.002
10^7	1.564	0.002

Table 2: Simulation results for the $M/M/2$ reduction with $\lambda = 0.6$ and $\mu = 1.0$, evaluated over increasing simulation horizons.

The $M/M/2$ case shows a similar convergence pattern. Early simulations overestimate the theoretical mean waiting time ($\mathbb{E}[W] = 1.5625$) due to initial bias and high variance. As the simulation length increases, the results stabilize and match the theoretical value. Confirming that the reduced mldel shows $M/M/2$ dynamics.

4 Long Term Statistics

4.1 Computed Metrics

Throughout the simulation, we track the following metrics. First, we measure the total waiting time, which is the elapsed time from a request’s arrival until its service is completed. Second, we measure the average queue length across all districts. Third, we calculate truck utilization, defined as the fraction of time each truck actively services a request. Finally, we record the rerouting rate, the number of rerouting events per unit time.

4.2 Batch Means Method

4.2.1 Warm-up Time

First, we need to settle on a warm-up time W . We can do this by first setting $WU = 0$, and visually check when the aggregate observations become stable. For the default values, this was effectively immediate. We think that this is caused by the small average queue length (around 0.19). The queues take up very little time to fill up to their average length, which is why the warm-up time for the system is so low. For this reason, we choose to set $WU = 0$.

4.2.2 Ensuring Sufficient Precision

We choose significance level $\alpha = 0.05$ and precision $\delta = 0.01$ (allowing for 1% relative error). Let r be the number of batches, with aggregate observations $\bar{X}_1, \dots, \bar{X}_r$. We compute the point estimate with $\hat{X}(r) = \frac{1}{r} \sum_{k=1}^r \bar{X}_k$. To ensure precision δ , r should satisfy the following inequality

$$r \geq \frac{t_{r-1, 0.975}^2 S^2(r)}{(\frac{\delta}{1+\delta} \hat{X}(r))^2} \quad (3)$$

with the 0.975-percentile of the Student’s t -distribution $t_{r-1, 0.975}^2$ and the batch means sample variance $S^2(r) = \frac{1}{r-1} \sum_{k=1}^r (\bar{X}_k - \hat{X}(r))^2$.

In our analysis, we used $r = 50$ and batch length $L = 5 \cdot 10^5$. As seen in Table 3, this suffices for the given precision and significance. Note that the Queue Length gives the tightest inequality. This can be explained by its large relative variance between batches. We left the Rerouting Rate outside of this analysis. Rerouting events are very rare, making its variance relatively big.

This drives up the number of batches we need to an unrealistic number, so the current precision for this statistic is low.

Statistic	Lower Bound
Waiting Time	0.8737
Queue Length	38.634
Truck 1 Utilisation	1.5532
Truck 2 Utilisation	1.2047
Truck 3 Utilisation	1.4879
<i>Rerouting Rate</i>	(55431)

Table 3: Lower Bound for each Statistic with $r = 50$ and $L = 5 \cdot 10^5$

4.3 Regenerative Method

To formalize the regenerative analysis, we define the system notation in Table 4.

Table 4: Model Notation and Variable Definitions

Variable	Description	Range
N	Total number of regeneration cycles	$\mathbb{Z}^+$
N_k	Total arrivals prior to the start of the k -th cycle	$\mathbb{Z}^+$
M_k	Number of requests served during the k -th cycle ($N_{k+1} - N_k$)	$\mathbb{Z}^+$
W_n	Waiting time of the n -th request	$\mathbb{R}^+$
R_n	Indicator if the n -th request involves a reroute (1 = yes, 0 = no)	$\{0, 1\}$
T_{N_k}	Continuous time epoch where a customer arrives at an empty system	$\mathbb{R}^+$
C_k	Continuous time duration of cycle k ($T_{N_{k+1}} - T_{N_k}$)	$\mathbb{R}^+$
$L(t)$	Queue length at time t	$\mathbb{Z}^*$
$U_i(t)$	Status of truck i at time t (1 = busy, 0 = idle)	$\{0, 1\}$

The regenerative method simulates N independent cycles. We look at two types of metrics: a Discrete time (request average) case and a continuous time (time average) case. In both cases, the simulation produces independent samples (X_k, Y_k) for each cycle $k = 1, 2, \dots, N$. To estimate the long-run properties, we apply the classic ratio estimator:

$$\hat{\theta} = \frac{\overline{X}}{\overline{Y}} = \frac{\frac{1}{N} \sum_{k=1}^N X_k}{\frac{1}{N} \sum_{k=1}^N Y_k} \quad (4)$$

For metrics averaged over discrete events (such as individual requests), X_k represents the total amount of events over all arrivals within cycle k , while $Y_k = M_k$ denotes the total number of requests served during that same cycle.

$$X_k = \sum_{n=N_k}^{N_{k+1}-1} Z_n, \quad Y_k = M_k = N_{k+1} - N_k \quad (5)$$

where Z_n is the value of the metric at the n -th request.

For time dependent metrics, X_k is obtained by integrating $Z(t)$ over the entire duration of cycle k , while $Y_k = C_k$ represents the total length of that cycle.

$$X_k = \int_{T_{N_k}}^{T_{N_{k+1}}} Z(t) dt, \quad Y_k = C_k = T_{N_{k+1}} - T_{N_k} \quad (6)$$

Regardless of whether the metric is discrete or continuous, an approximate $100(1-\alpha)$ confidence interval for the ratio estimator $\hat{\theta}$ is reached via:

$$\hat{\theta} \pm t_{N-1, 1-\frac{\alpha}{2}} \sqrt{\frac{\widehat{\text{var}}(V)}{\bar{Y}^2 N}} \quad (7)$$

where

$$\widehat{\text{var}}(V) = \widehat{\text{var}}(X) + \hat{\theta}^2 \widehat{\text{var}}(Y) - 2\hat{\theta} \widehat{\text{cov}}(X, Y) \quad (8)$$

4.3.1 Application to Metrics

We use this method to determine the following metrics:

- **Waiting Time** (Discrete time Case): Achieved by setting $Z_n = W_n$. Here, $X_k = W_k^{\text{total}} = \sum_{n=N_k}^{N_{k+1}-1} W_n$ represents the cumulative waiting time of all requests in cycle k , and $Y_k = M_k$.
- **Rerouting Rate** (Discrete time Case): Because this is defined as the fraction of reroutes against the total amount of requests, we get this by setting $Z_n = R_n$. Here, $X_k = R_k^{\text{total}} = \sum_{n=N_k}^{N_{k+1}-1} R_n$ represents the total number of rerouting events triggered during cycle k , and $Y_k = M_k$.
- **Queue Length** (Continuous time Case): Achieved by setting $Z(t) = L(t)$. $X_k = L_k^{\text{total}} = \int_{T_{N_k}}^{T_{N_{k+1}}} L(t) dt$ represents the queue length, and $Y_k = C_k$.
- **Truck Utilization** (Continuous time Case): Achieved by setting $Z(t) = U_i(t)$. Here, $X_k = U_{i,k}^{\text{total}} = \int_{T_{N_k}}^{T_{N_{k+1}}} U_i(t) dt$ represents the total active duration for truck i within the cycle, and $Y_k = C_k$.

5 Results

5.1 Batch Means Method

The final point estimations and 95% confidence intervals using the Batch Means Method with $r = 50$ and $L = 5 \cdot 10^5$ using random seed 70 are given in Table 5. Note that only the Rerouting Rate does not meet precision $\delta = 0.01$.

Table 5: Final Results of Batch Means Method with precision $\delta = 0.01$

Statistic	Point Estimation	95% CI
Waiting Time	1.4241	[1.4223, 1.4260]
Queue Length	0.3102	[0.3083, 0.3122]
Truck 1 Utilization	0.4664	[0.4655, 0.4673]
Truck 2 Utilization	0.4665	[0.4657, 0.4672]
Truck 3 Utilization	0.4661	[0.4654, 0.4669]
Rerouting Rate	$2.080 \cdot 10^{-5}$	$[0.1506 \cdot 10^{-5}, 2.654 \cdot 10^{-5}]$

5.2 Regenerative Method

The final results of the Regenerative Method are given in Table 6.

Table 6: Final Results of Regenerative Method

Statistic	95% CI
Waiting Time	[1.4193, 1.4304]
Queue Length	[0.3050, 0.3154]
Truck 1 Utilization	[0.4627, 0.4670]
Truck 2 Utilization	[0.4634, 0.4677]
Truck 3 Utilization	[0.4643, 0.4686]
Rerouting Rate	$[0.1175 \cdot 10^{-5}, 2.9622 \cdot 10^{-5}]$

Apparently, the Batch Means Method and the Regenerative Method give very similar results.

5.3 Friendliness Comparison

We use the Batch Means Method to compare the point estimations using friendliness values $p = 0.5$ and $p = 0.51$. This comparison is displayed in Table 7.

Table 7: Comparison of Statistics between $p = 0.50$ and $p = 0.51$

Statistic	$p = 0.50$	$p = 0.51$	Relative Difference
Waiting Time	1.4241	1.4218	-0.16%
Queue Length	0.3102	0.3062	-1.29%
Truck 1 Utilization	0.4664	0.4670	0.13%
Truck 2 Utilization	0.4665	0.4670	0.11%
Truck 3 Utilization	0.4661	0.4666	0.11%
Rerouting Rate	$2.080 \cdot 10^{-5}$	$1.560 \cdot 10^{-5}$	-25.00%

As we expected, altering the friendliness value does not result in big differences in the statistics.